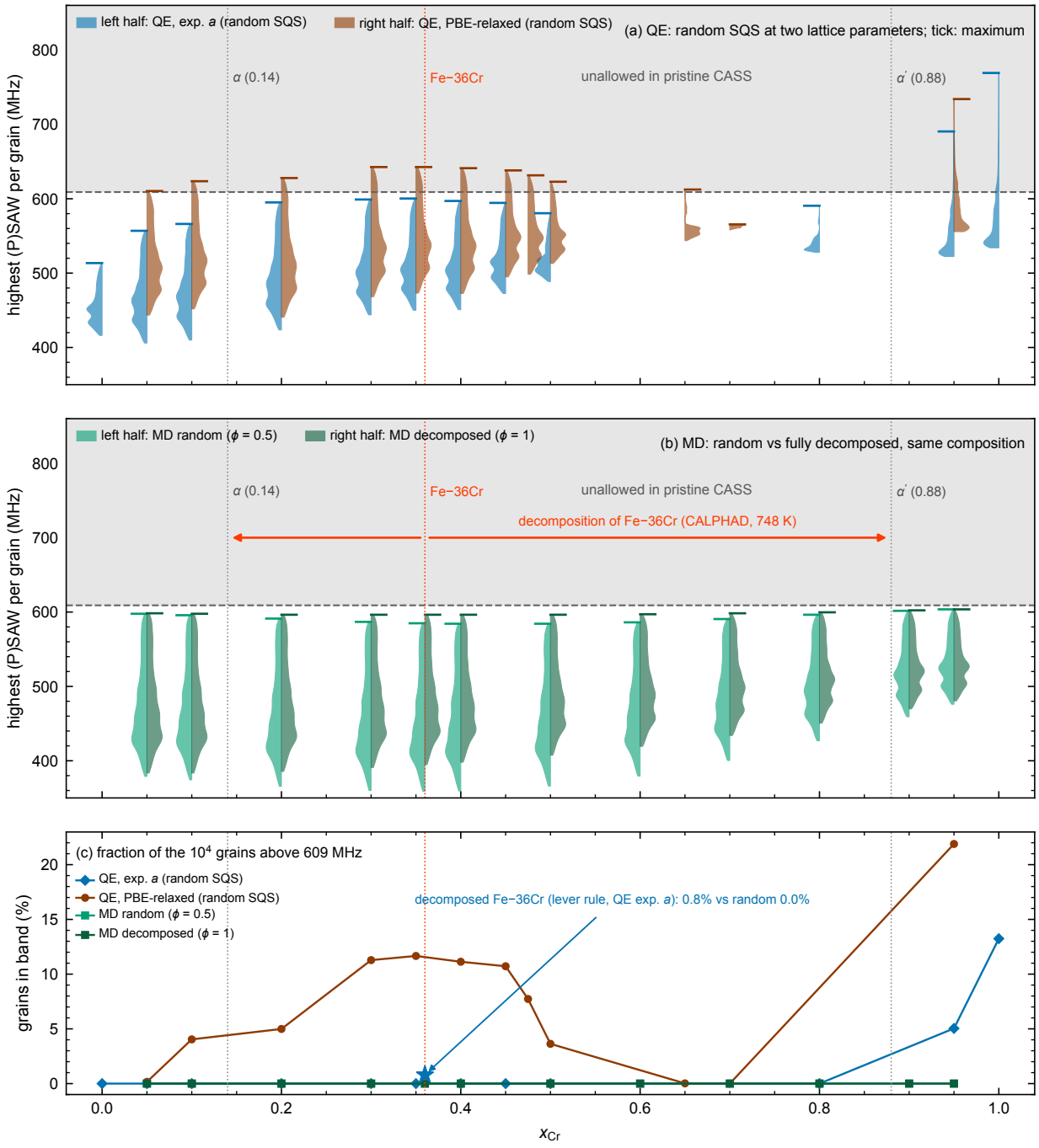

left half: QE, exp. a (random SQS)
right half: QE, PBE-relaxed (random SQS)
(a) QE: random SQS at two lattice parameters; tick: maximum
α (0.14)
Fe−36Cr
unallowed in pristine CASS
α′ (0.88)
highest (P)SAW per grain (MHz)
800
700
600
500
400
left half: MD random (ϕ = 0.5)
right half: MD decomposed (ϕ = 1)
(b) MD: random vs fully decomposed, same composition
decomposition of Fe−36Cr (CALPHAD, 748 K)
(c) fraction of the 10^4 grains above 609 MHz
QE, exp. a (random SQS)
QE, PBE-relaxed (random SQS)
MD random (ϕ = 0.5)
MD decomposed (ϕ = 1)
decomposed Fe−36Cr (lever rule, QE exp. a): 0.8% vs random 0.0%
grains in band (%)
20
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
x_Cr